

<https://doi.org/10.1038/s43247-024-01768-y>

Microplastics deposition in Arctic sediments of Greenland increases significantly after 1950



Karla Parga Martínez ^{1,4}✉, Thorbjørn J. Andersen ², Vitor da Silva³, Jakob Strand³ & Nicole R. Posth¹

Marine sediments are archives of environmental change with pollutants potentially acting as chronographic markers of the Anthropocene. Particularly, the vertical transport and burial, as well as slow degradation rate of microplastics, indicate an eventual incorporation into the geological record. A high-resolution reconstruction of microplastics records requires high sedimentation rates, and in Disko Bay (Greenland), this quality coincides with a need for plastic pollution data. Here, a Greenlandic sediment core dated back to 1930 ± 2 was reconstructed for microplastics accumulation via micro-Fourier transform infrared spectroscopy. We show 85 years of fluctuating microplastics accumulation ($987 - 16,645$ particles Kg^{-1}) down to $20 \mu\text{m}$ and diversified into eight polymers. Polyethylene (47%) and polypropylene (32%) were more abundantly present through time. Microplastics accumulation increases significantly after 1950 along with major socio-economic development in the area, suggesting an influence from regional stressors. Regional microplastics reconstructions must therefore be considered in the pursuit of an Anthropocene global plastic horizon.

Since the Great Acceleration (mid-20th century), human activities have left significant changes in the Earth System¹, a footprint of which can be traced in suitable depositional environments. Current chronostratigraphic markers mainly occur within marine sediments where proxy signals are well preserved and continuously recorded². Sediments are contaminant sinks, and microplastics (MP) accumulation in the strata is no exception. The global distribution, long residence time and durability of plastics indicate their potential suitability as auxiliary markers of the Anthropocene³. Plastic influx to the seafloor is incorporated into the sediment leading to a potential footprint of mismanaged plastic through time. This footprint, however, may be regional, rather than global, as plastics do not all follow the same transportation pathways or degradation rates due to their diverse size, morphology and chemical composition⁴. Hence, the reconstruction of MP records should consider regional influence on deposition and accumulation patterns of plastics. This is particularly relevant in areas of scarce data for which sources and pathways may also be underestimated⁵. Tracking a MP footprint first requires a reliable chronology, normally established via ^{210}Pb (lead) for the recent period from the 1920s to present⁶. However, fine-sediment environments are often difficult to date as bioturbation, physical action (e.g. ocean currents) and human action (e.g. bottom trawling fisheries) disturb the chronology, all of which must be considered in age models. Records of MP

accumulation in dated cores from freshwater, shallow marine sediments^{7–11}, and the deep sea^{12–16} suggest MP as auxiliary markers of the Anthropocene.

Although still considered pristine, the Arctic might be more susceptible to MP accumulation in strata. In this cold-weather region, waste management systems are challenged by limited infrastructure, sparse and isolated coastal settlements, as well as harsh weather conditions. Indeed, reports of MP in snow, sea ice, water column and surface sediments^{17–20} from across the Arctic continue to increase. However, there are no MP accumulation records in Arctic sediments. Yet, an anthropogenic footprint might have already imprinted the sedimentary record, potentially recording the ongoing shift towards improved waste management systems. In Greenland, traditional waste management practices resort to incineration, landfilling, waste export to Denmark²¹ and direct runoff of wastewater to the sea²². Recent attempts to centralize solid waste collection from small settlements to the incineration facilities in Nuuk and Sisimiut have been implemented²³. However, collection centers still face solid waste dispersal during frequent storm events. These current practices along with reports from beach litter surveys^{24–26} suggest that (macro)plastic pollution in Greenland may largely be local rather than long-range transport.

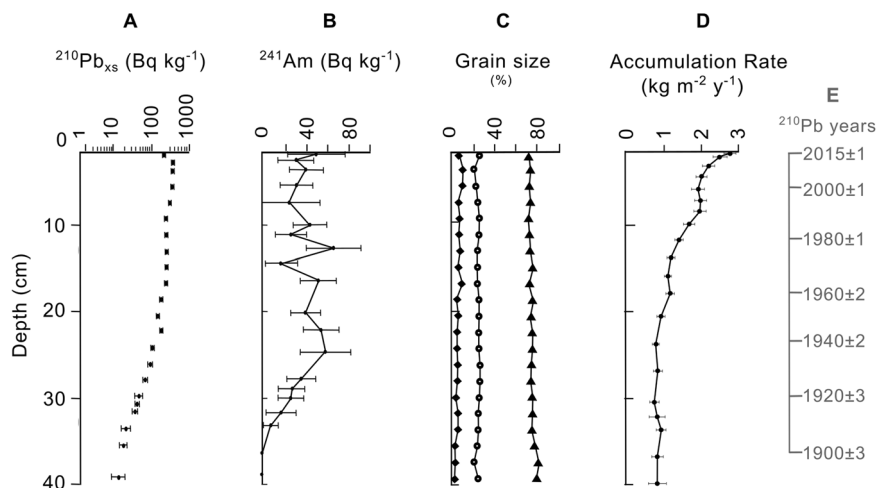
Greenland is the largest island in the world holding the lowest population (57,000 inhabitants) as 81% of the territory is covered by ice. The

¹Section of Geology - Department of Geosciences and Natural Resource Management (IGN), University of Copenhagen, Copenhagen, Denmark. ²Section of Geography - Department of Geosciences and Natural Resource Management (IGN), University of Copenhagen, Copenhagen, Denmark. ³Department of Ecoscience, Aarhus University, Roskilde, Denmark. ⁴Present address: Department of Chemical Engineering, McGill University, Montreal, QC, Canada.

✉ e-mail: karla.pargamartinez@mcgill.ca

Fig. 1 | Sediment analysis and Radiometric data.

A Exponential decay of unsupported ^{210}Pb showed via excess $^{210}\text{Pb}_{\text{xs}}$. **B** Activity of ^{241}Am displaying two global (peaks) time markers as validation of ^{210}Pb dating. **C** Granulometric composition in percentage of clay (circles), silt (triangles) and sand (diamonds). **D** Sediment Accumulation Rate. **E** Estimated years based on ^{210}Pb dating. Error bars are shown in panels **A**, **B** and **D**.



population lives in coastal settlements concentrated southwest²⁷, among the largest Ilulissat and Asiaat are located around Disko Bay just above the Arctic Circle. Disko Bay, as the rest of the country, largely relies on coastal and offshore fisheries (1/3 of the revenue mainly stands from Greenlandic halibut and Arctic prawn fisheries²⁷). Tourism is the best-developed market with a domestic airport in Ilulissat, a high number of tourism operators and, seasonal cruise ships²⁷. East of Disko Bay, the glacier Sermeq Kujalleq discharges 10% of the total iceberg production in Greenland directly into the Bay²⁸ along with meltwater and glacial marine sediments; the latter contributing to high sedimentation rates. Considering all the above, environmental, social and economic stressors might contribute to local plastic release in the area. Reports of MP pollution in Greenland remain scarce^{29–34}, even more so for MP in sediment^{17,35}. In this article, a MP pollution archive was reconstructed in glacial marine sediment from Egedesminde Dyb, Disko Bay to search for a footprint of the Anthropocene. Microplastics deposition in Arctic sediments of Greenland increased significantly after 1950, and the regional influence from potential sources of plastic pollution is discussed to determine the suitability of reconstructing MP records and their use in the pursuit of Anthropocene markers.

Results and discussion

Glacial marine sediment analysis and chronology

The glacial marine sediment core spanned from 1860 to 2015, which bridges the Plastic Boom of the 1950s (here 24 cm depth). Surface contents of unsupported ^{210}Pb around 400 Bq Kg⁻¹ (Becquerel per kilogram) showed a clear tendency for exponential decline in the core. The calculated flux of unsupported ^{210}Pb was 1,035 Bq m⁻² y⁻¹ (Fig. 1A), which is around an order of magnitude higher than the expected atmospheric flux³⁶. This indicates that the site is subject to intense sediment focusing with a high sediment accumulation rate (mean 1.36 Kg m⁻² y⁻¹ Fig. 1D). Here, an average sedimentation rate since 1930 (0.34 ± 0.01 cm y⁻¹) allowed for a triennial to decadal resolution of the chronology. Disko Bay is subject to glacial marine processes from the glacier Sermeq Kujalleq melt-water, and sedimentation rates can vary from 1.7 cm y⁻¹ in ice-proximal glacial marine mud to ~ 0.007 –0.5 cm y⁻¹ for overlying distal sediments³⁷. However, other deep-sea locations hold lower sedimentation rates, so that sediment deposited in the 1950's can be found as shallow as 4 cm depth (9×10^{-4} – 5.5×10^{-2} cm y⁻¹ sedimentation rate¹³). Therefore, sedimentation rate is the main determinant of chronology resolution and is a limiting factor in the reconstruction of MP archives in sediments of the Anthropocene.

The influence of Sermeq Kujalleq glacier on the sedimentary system in Disko Bay can also be seen in the granulometric analysis. Fine sediment (22% clay, 75% silt and 3% sand, Fig. 1C) homogeneously dominated the grain size distribution throughout the core. The porosity and water content significantly decreased with depth (Spearman's ρ , $r > 0.72$, $p < 0.002$, Supplementary Fig. 1 and 2) consistent with sediment compaction. Sediment

mixing potentially disturbing the chronology and MP accumulation did not appear to be significant based on the radiometric profiles, bioturbation and organic matter (OM) content. The profiles of ^{210}Pb and ^{241}Am (Americium) did not show significant mixed radioisotopes measurements; only traces of ^{241}Am were found in layers dated before 1940, indicating that some but limited mixing is occurring (Fig. 1B). Traces of bioturbation were not observed in the upper layers nor throughout the core, further supported by the low content of OM based on total organic carbon (TOC mean 2.32%) and total organic nitrogen (TON mean 0.28%) (Supplementary Fig. 3). Organic matter is a limiting factor for benthic biomass and bioturbation activity³⁸. Generally, the Arctic Ocean has low content of OM (2%) and low benthic biomass in the deep ocean, which results in low sediment mixing rates with reported bioturbative transports between 0.01 and 0.11 cm⁻¹ y⁻¹ and 0.2 cm⁻¹ y⁻¹^{38,39}. Thus, some bioturbation may occur in Egedesminde Dyb, but it is not significant at 850 m water depth.

MP accumulation in the Anthropocene

The MP burial rate fluctuated between 785 and 18,429 particles m⁻² y⁻¹ and peaked in 1967. While it significantly increased after the Plastic Boom in 1950 (Wilcoxon: $Z = -2.525$ $p = 0.011$) (Fig. 2), MP burial rates above 10,000 particles m⁻² y⁻¹ were only seen in recent years (2000, 2012 and 2015). Despite the MP burial rate being significantly influenced by the sedimentation rate (Spearman's ρ , $r = 0.57$, $p = 0.03$), the MP distribution profile here and particularly the peak of 1967 seems unlikely to result from MP downward migration. MP behavior in the sediment compartment is still poorly understood and while MP particle movement might occur at some extent, from a sedimentological perspective, the micropores in silt and clay-rich sediments limit air and water movement, which would limit MP movement in turn. Further, the data distribution in MP accumulation, polymer and size variability (Supplementary Figs. 4 and 5) did not show any indication of MP particle downward migration that might be displacing MP towards older layers. Thus, the abrupt peak of 1967 might be an anomaly as such high quantities were not recorded in the following years. A possible explanation is the influence of the chronology resolution as more time points between 1960 and 1974 could have allowed a more detailed reconstruction of the MP record. Unfortunately, sediment cores limit the material available for analyzes at the expense of building reliable reconstructions of contamination archives². Here, concerns about sample size for MP extraction led to pooling two samples together, which reduced the age model resolution. Ideally, multiple sediment cores should be analyzed to enhance the comprehensiveness of quantifying MP pollution in the area. However, this study was conducted in an archived sample from 2015 and the number of available sediment cores suitable for dating and MP analysis was a limiting factor to increase sample representativeness of Disko Bay.

Environmental variability of sedimentary systems may also influence the variability in MP accumulation; especially Disko Bay is subject to glacier-

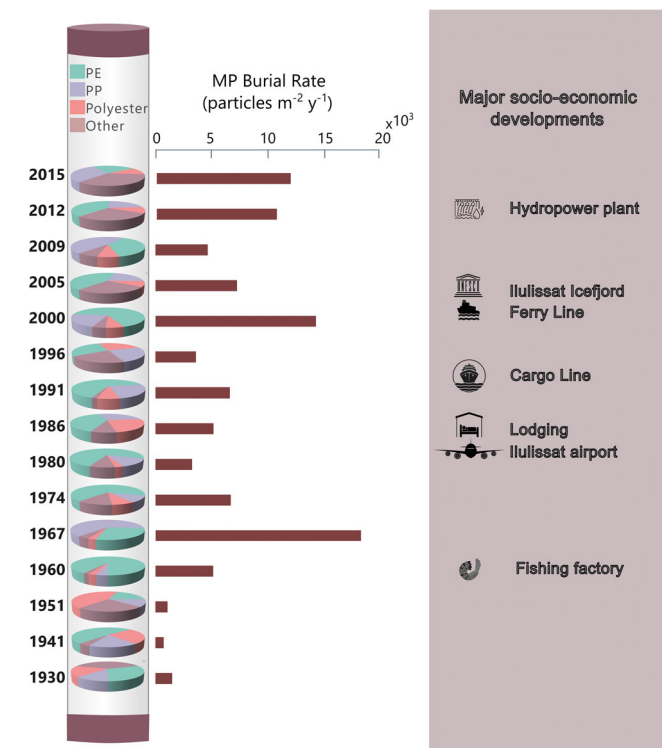


Fig. 2 | Microplastics burial rate in glaciarmine sediment from 1930 to 2015 and potential anthropogenic sources. MP accumulation significantly increased after the 1950s Plastic boom and diversified into eight polymer types. PE, PP and Polyester were constantly present. Other refers to PC, PS, PVC, PU and PMMA. Major socioeconomic development in Ilulissat (Disko Bay) are shown to the right: the fishing factory opening in 1961; Ilulissat airport operations started in 1984 and shortly after lodging was opened to the public; cargo lines traverse the Bay since 1993; the Ferry Line has expanded their fleet since first appearance in 2004; Ilulissat Icefjord is declared a UNESCO site in 2004; the hydropower plant has been active since 2012.

driven sedimentary processes from glacier discharge and iceberg drift⁴⁰. Meltwater transports airborne MP particles deposited on the ice⁴¹ and drifting icebergs might release frozen legacy MP particles as reported from sea ice and glacial ice^{42–44}. In 85 years of accumulation, a total of 1,064 particles were found in the glaciarmine sediment core. MP accumulation ranged between 987.4 and 16,645 particles Kg^{-1} (mean $\pm$ SD 4,272.3 $\pm$ 3854.4) and the estimated mass ranged between 0.03 and 0.685 mg Kg^{-1} (Supplementary Table 1). The trend fluctuated through the years with no clear increase towards present time. However, the pre-1950s accumulation was significantly lower (Wilcoxon $Z = -2.52$, $p = 0.012$) and in the post-1950s period, the peak of 1967 reached 16,645 particles Kg^{-1} (Fig. 2). Along with environmental variability, anthropogenic activities could also explain the fluctuation in MP accumulation. For instance, with the start of the Great Acceleration and the Plastic Boom, major socio-economic activities flourished in Disko Bay. The historical use of plastic in any anthropogenic activity can be a potential source of MP pollution due to historical landfilling of solid waste and ongoing direct discharge of wastewater into the sea²¹.

Potential anthropogenic sources of MP pollution derived from socio-economic development in the area include the establishment of a fishing factory by the port (1961) for the use of fishing nets and packaging material, the construction of Ilulissat domestic airport (1984) boosting tourism, inevitably leaving a footprint in the environment and increasing the demand for lodging and shipping. Maritime transport is considered a major source of marine pollution leading to the International Convention for the Prevention of Pollution from Ships (MARPOL Annex V) ban on dumping waste and specifically plastic at sea. Further, the Icefjord UNESCO site (2004) led to the creation of the Icefjord Centre (2021) next to the archeological site

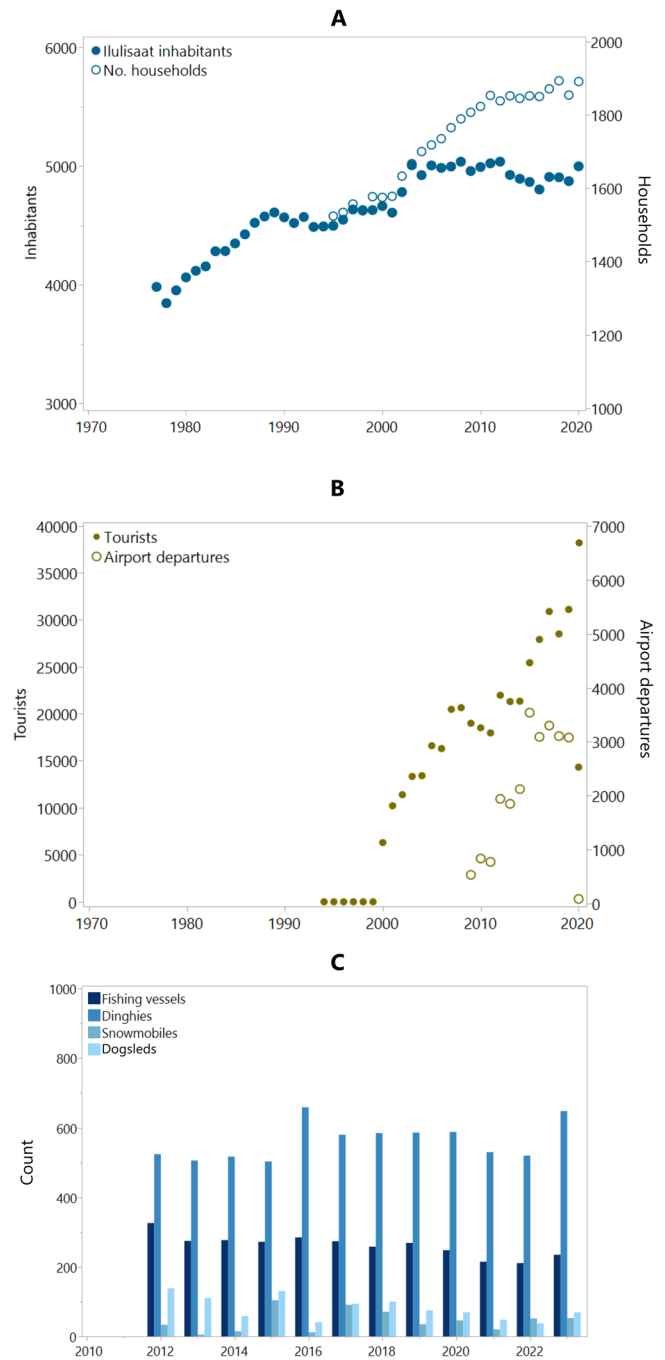


Fig. 3 | Anthropogenic stressors in Disko Bay. A Population growth and number of households in Ilulissat. B Number of tourists and local airport departures. C Fishing effort in number of vessels. Source: Statistics Greenland <https://bank.stat.gl/pxweb/en/Greenland> and Greenland's Fisheries License Control Authority (GFLK).

Sermermiut. The Icefjord attracts around 25,000 visitors yearly⁴⁵ exerting an anthropogenic pressure to ecosystems (e.g. littering), and to waste management in a geographically remote and environmentally challenged region. The hydropower plant built near the bay (2012) could act as a transportation pathway and reintroduce MP from ice and lake water supply. The hydropower plant supplies electricity, drinking water and heating to the growing town and tourism demand (126% population growth since 1977, 128% growth in households' number since 1994, tripled airport departures since 2009 and quintupled tourist stays since 2000) (Fig. 3). All these socio-economic activities are potential local sources of MP pollution for the accidental or intentional leaks to the environment during daily activities,

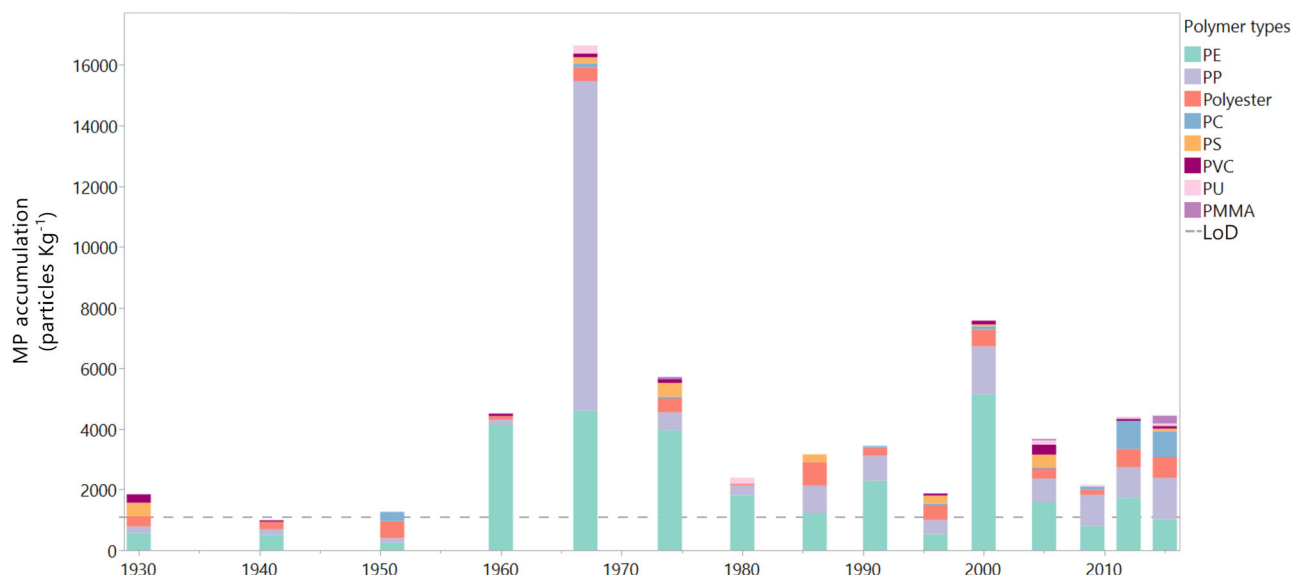


Fig. 4 | Microplastics accumulation and polymers distribution throughout the years. The most abundant polymers constantly present across time were PE, PP and Polyester. No increasing trends towards present time were detected. Dotted line refers to LoD.

and because of the general lack of efficient waste management systems in the Arctic.

Plastic as a new material of global accumulation has been suggested as a potential stratigraphic indicator of the Anthropocene⁴⁶. As such, records of MP accumulation have been reconstructed from freshwater and shelf sediments showing a general MP increase towards the present time^{7,8,12,15,16,47}. In the deep sea, MP accumulation over time has been reported between 197 particles Kg⁻¹ and 6,939 particles Kg⁻¹^{13,48} and is even higher at abyssal depths (997,800 particles Kg⁻¹)¹⁴. A complete overview of historical records of MP in aquatic sediment, which includes a sedimentological profile can be found in Supplementary Table 2. MP reports in Arctic sediment are particularly variable, from absent or low quantities in the Chukchi Sea and Svalbard fjords^{49,50} to 13,331 particles Kg⁻¹ in the Fram Strait^{18,35}. Latest MP reports from the Pacific Arctic and Siberia show almost 2,000 particles Kg⁻¹^{51,52}. On the one hand, MP pollution shown here from Disko Bay, is similar to reports from Arctic sediments, and as expected, the MP accumulation was low in the pre-Plastic Boom period. On the other hand, the fluctuating post-Plastic Boom trend did not resemble any increase towards the present time nor followed the global plastic production curve as commonly correlated but instead regional factors, both environmental and anthropogenic, seemed to have more strongly influenced MP accumulation in Disko Bay. Small water bodies, such as lakes or bays, are more susceptible to local sources of MP pollution and atmospheric deposition rather than expansive sources carrying a MP global trend⁵³. Physical, geological and biological local characteristics, as well as human stressors, influence the reconstruction and interpretation of MP records, i.e., potential disruptors of the chronology (e.g., bioturbation activity, bottom-trawl fisheries, deep-sea mining) and anomalies in MP records (e.g., plastic accumulation zones, sewage outlets). Thus, regional factors of MP sources and transportation pathways must be considered in the pursuit of a global plastic horizon of the Anthropocene⁴⁶.

Regional stressors in Disko Bay include glacier-driven processes, windstorms that disperse plastic items from landfills and collection facilities²⁵, tourism and, fisheries including common practices, such as rope cut-offs discarded at sea during net mending²⁴. Data from beach litter surveys suggest plastic items are of local origin as they usually appear intact and non-degraded^{24,25,54}. Here, the term local sources refers to locally disposed, not necessarily by the locals but also by transient travelers. Current efforts to improve waste management systems aim to mitigate anthropogenic stressors^{22,55}; however, unpredictable Arctic weather and harsh terrain

challenge the implementation of traditional waste management practices e.g. waste segregation, recycling, bio-waste for compost.

Polymeric composition

While it is impossible to know the exact source of MP in a depositional setting, the polymeric composition of an MP sample could indicate a product or industry. Here, eight polymer types were identified, the most abundant polyethylene (PE) and polypropylene (PP) accounted for 79% of total particles (47% and 32%, respectively), followed by polyester (PET) accounting for 10%. These three plastic contributors were constantly present throughout the years. Polycarbonate (PC), polystyrene (PS) and polyvinylchloride (PVC) were present in less than 5%, and polyurethane (PU) and polymethyl methacrylate (PMMA) were present in less than 1% (Fig. 4). The overall polymer diversity (Shannon-Wiener diversity index $H' = 2.45$) did not significantly increase with time (Spearman's ρ , $r = 0.2643$, $p = 0.3412$) although it did increase in the last decade reaching 1.69 (H') in 2015 with presence of all eight polymer types.

Among the most consumed and produced plastics, PE and PP are commonly found in the marine environment and the seafloor⁵⁶. They are widely used in household products, construction and transportation, which are the largest industries in Greenland after fisheries²⁷. Regionally, commercial waste is larger than household waste (2,224 tons vs 1,177 tons in 2022, Receiving Station weight Ilulissat Incineration Plant, personal communication) and it is currently divided into 21 categories (Supplementary Table 3). Fisheries are diversified into fishing vessels, dinghy boats, dog sleds and snowmobiles amounting to 1005 actively fishing in Disko Bay (GFLK, personal communication). Key fishing waste items include PP nets, ropes and lines accounting for 5870 Kg in 2022 (Supplementary Table 3), and already categorized in the OSPAR Marine Litter RAP Action 35⁵⁷. Polyester is also one of the main plastic waste contributors. PET is extensively used in the beverage industry for which a deposit scheme was instituted in West Greenland in 2002 showing 90% returned success⁵⁸. However, this deposit scheme does not include water bottles, which likely end in an uncontrolled landfill awaiting incineration²³. Primary and secondary MP interact with the environment and can undergo multiple transportation pathways. Microplastics can remain in suspension, be ingested by biota, transported to land, deposited on shorelines, or nanofragment, but their ultimate fate is the seafloor^{59–61}. The downward transport of MP in the water column is facilitated by marine snow, biofouling and flocculation^{62–64}.

One of the greatest challenges in reliable MP quantification in the environment is contamination during sampling and laboratory analysis.



Fig. 5 | Microplastics particle size and morphology. Size distribution pre- and post-1950 was dominated by small particles (20–100 µm) whereas large particles (> 100 µm) were almost exclusively present after 1950. MP particle morphology

based on the length-to-diameter ratio (major to minor dimensions) showed 85% were fragment-like particles. Long fiber-like particles were mainly composed of PET.

Airborne MP and fibers surrounding the environment can cause overestimation of MP counts from the field and laboratory sample processing. As a precautionary measure, strict contamination protocols are part of Quality Assurance and Quality Control (QA/QC). The use of blanks throughout sample collection, processing and analysis is required to establish the Limit of detection (LoD), which reflects the efficiency of the precautionary methods. Field blanks can account for airborne MP and fibers during sampling, from either the environment or plastic materials from sampling gear and equipment, which cannot always be avoided. For instance, the glaci-marine core was collected in a PC tube, which could have produced PC particles from abrasion. Based on the intermittent PC distribution and low quantities across time, it does not seem to be the case. However, PC particles increased after 2012 (Fig. 4) with the most recent layer (2015) also being the most exposed to airborne microplastics during sampling. Unfortunately, the use of an archived glaci-marine core in this study did not allow the implementation of field blanks during sampling. Thus, contamination from airborne MP to the topmost sample is possible to have occurred although MP accumulation did not significantly increase in recent years (Spearman's ρ , $r = 0.335$, $p = 0.221$).

Procedural blanks in the laboratory showed the presence of five polymers: PE, PP, Polyester, PS and PU. The overall LoD was 77 particles, mainly from PE for which some of the PE accumulation fell under the LoD (Supplementary Table 4). The presence of MP particles in the procedural blanks shows the complexity of accurately quantifying environmental MP and the uncertainties in MP studies⁶⁵.

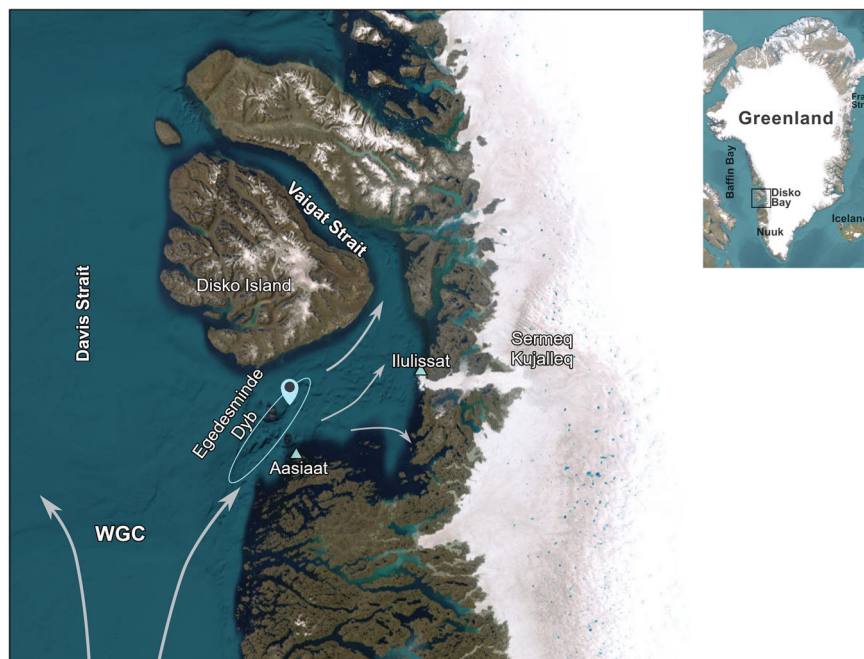
Size distribution and MP morphology

MP accumulation was dominated by small particles (85% between 20 and 100 µm, $\bar{x} = 73$ µm), which greatly overlap in size with silt grains (2–63 µm, 75% silt composition). Hence, the importance of removing the inorganic fraction to obtain an exhaustive footprint of MP accumulation, within the

FTIR limit of detection. To this end, the method for fine-grained sediments applied to the glaci-marine samples allowed the inclusion of the smallest size classes (< 100 µm), which are otherwise methodologically overlooked and underestimated⁶⁶. The abundance of MP particles is known to rapidly increase when particles become smaller than 100 µm⁶⁷; for instance, 99% of MP in sediments from the Fram Strait were < 150 µm⁶⁸. Concerns related to small MP in the marine environment include particle uptake by organisms, interaction with microorganisms, food web bioaccumulation, chemical toxicity, e.g. absorption, leaching and biogeochemical cycling^{69,70}. Only a few particles among the large fraction (> 100 µm) found in this study exceeded 500 µm. There was no increasing trend in size towards present time in the large fraction nor a decreasing trend towards pre-1950s in the small fraction that would indicate fragmentation at the sink (Supplementary Fig. 4). This might suggest that MP already enters Disko Bay at a small size. In this study, the majority of MP particles were < 50 µm (47%), which limits visual evaluation to acquire physical information such as particle morphology and color. MP morphology can be broadly classified into fiber-like particles based on a length to diameter ratio > 3 (major and minor dimensions) and fragment-like particles with a ratio ≤ 3, as recommended by the European Chemical Agency (ECHA)⁷¹ (Supplementary Fig. 6). Here, fragment-like particles were more abundant (85%) than fiber-like particles (15%). In terms of size, small fragment-like particles dominated the MP record (79%) whereas fiber-like particles size was almost equally distributed (46% vs 54%, small vs large particles). However, all polymer types were present in both size fractions and morphology although Polyester dominated the composition of large fiber-like particles (Fig. 5).

It should be noted that particle morphology obtained through spectral mapping carries a degree of uncertainty. Specifically, it may not fully account for the influence of spectral measurements on particle edges and scattering, potentially affecting the particle area assigned for each MP. Furthermore, the presence of surrounding particles and the three-

Fig. 6 | Map of Disko Bay and sampling site at Egedesminde Dyb. Two of the largest settlements in Greenland, Aasiaat and Ilulissat (blue triangles) are located by the Bay. The fast-flowing glacier Sermeq Kujalleq drains the Greenlandic Ice Sheet through Ilulissat Icefjord. Schematic route of the West Greenland Current (WGC) indicated with gray arrows.



dimensional aspects of individual MP particles represent additional limiting factors that should be considered when applying morphological classification from spectral imaging.

Conclusions

A footprint of MP accumulation has been recorded in glacial marine strata of Disko Bay. This footprint seems to be influenced by regional signals from anthropogenic activities and special environmental features, which are reflected in the fluctuation of MP accumulation through time. This indicates that this seemingly pristine region in the Arctic Circle has therefore received a hidden influx of plastic pollution to the seafloor since 1930. The MP record differed in particle abundance, polymeric composition and size between pre- and post-1950 periods. PE and PP, however, were consistently the main plastic contributors. The predominance of small MP (20–100 μm) and the lack of fragmentation trends suggest that MP may arrive to the seafloor already at small sizes. However, these results only show a small section of a heterogeneous sedimentary system. It is therefore recommended to expand the number of cores to retrieve a comprehensive footprint of the Anthropocene in Greenland. In addition, the reconstruction of pollution archives, including MP, require (1) a reliable chronology preferably in high sedimentation environments to counterbalance the age model resolution and the amount of analyzed material, (2) the extraction method should target small MP (< 100 μm) to avoid underestimating the accumulation, and (3) blanks should always be included in the QA/QC protocols. The reconstruction of MP records must consider the influence of regional stressors in the pursuit of a global plastic horizon of the Anthropocene.

Methods

Study area

Disko Bay is located in West Greenland at 69° N latitude, it is approx. 100 km wide and bounded by Ilulissat Icefjord and mainland Greenland to the East, and Baffin Bay to the West (Fig. 6). It is characterized by a polar maritime climate with mean annual temperature of -4°C and precipitation of 450 mm⁷². The bathymetry of the Bay varies between 300 and 500 m except for Egedesminde Deep, which is a through deeper than 900 m⁴⁰. Oceanographic conditions are driven by seasonal ice cover, the West Greenland Current and freshwater from the Greenland Ice Sheet⁷³. The fastest-flowing glacier in Greenland, Sermeq Kujalleq, produces the largest icebergs in the world that flow through Ilulissat Icefjord⁴⁰ and transports glacial marine sediment mainly composed of silt and clay particles³⁷.

Sample collection

A glacial marine sediment core was collected in Disko Bay (69°03' 67" N, 53°03' 39" W) in August 2015 (Fig. 6). The site was selected due to the suitability to hold a chronology of the Anthropocene and the multiple human stressors on the Bay (e.g. human settlements, tourism, fisheries). The core was retrieved from 850 m water depth using a Ruhmor Lor corer (PC tube, 71 cm $\times$ 7 cm) and contained 20 cm of top water. The core was sealed, transported vertically by container to the institute, and stored vertically at 4 °C until further analysis.

All equipment was thoroughly rinsed with pre-filtered MilliQ water before and in between samples, the room was kept closed to avoid contamination. The sediment core was sliced with a stainless-steel knife in 1-cm horizons divided equally in two halves. One half was collected in a PE bag for radionuclide dating, and the other half was collected in a glass jar (Refer to Quality Control), sealed with tinfoil and stored for MP analysis at room temperature.

Radionuclide dating

The samples from the glacial marine core and the upper 50 cm of a sister core (used only for chronology validation and sliced in the same manner as previously described) were freeze-dried and weighed to determine the dry bulk density. The samples were analyzed for the activity of ^{210}Pb , ^{226}Ra and ^{241}Am via gamma spectrometry at the Gamma Dating Center, (IGN) University of Copenhagen. The measurements were carried out on a Canberra ultralow-background Ge-detector. ^{210}Pb was measured via its gamma-peak at 46.5 keV, ^{226}Ra via the granddaughter ^{214}Pb (peaks at 295 and 352 keV) and ^{241}Am via its peak at 59.5 keV. Excess ^{210}Pb ($^{210}\text{Pb}_{\text{xs}}$) was determined by subtracting the specific activity of ^{214}Pb via ^{226}Ra from the total ^{210}Pb specific activity.

The specific activity of each sample and total uncertainties (1 SD) of calculated specific activity values are expressed in Becquerel per kilogram of dry weight (Bq kg^{-1}). The chronology was derived from a constant rate of supply model in which the inventory in the lowermost part of the cores was calculated based on an exponential regression of unsupported ^{210}Pb vs cumulative mass depth in the bottom part of the core.

Sediment analysis

After the chronology was established, a subsample of the same half layer was used to analyze the grain size composition using a Mastersizer 2000 laser diffraction particle analyzer (Malvern Panalytical, IGN). The results are

Table 1 | Microplastics particles found in the Air blanks of the cleanroom laboratory

Air Blank	1	2	3	4	5	6	7	Average	SD
PE	3	0	5	3	0	0	3	2	2
PP	0	1	0	1	4	0	0	0.86	1.46
PET	1	2	6	2	1	6	11	4.14	3.72
PMMA	0	0	0	0	0	0	1	0.14	0.38
Total	4	3	11	6	5	6	15	7.14	4.30

Table 2 | Summary of microplastics particles found in the Procedural blanks

	PE	PP	Polyester	PC	PS	PVC	PU	PMMA
Mean	9.6	10	2	0	0.6	0	0.3	0
SD	12.6	7.2	2.5	<1	1.1	<1	0.5	<1

shown in percentage of clay, silt and sand. The organic matter content was determined via TOC and TON measurements, the samples were HCl (Hydrochloric acid) fumigated prior to Carbon to Nitrogen (C:N) analysis to remove calcites. The organic matter content is shown in terms of C:N ratio. The water content and porosity of the sediment core across the depth profile were calculated applying the formulas from Courteney-Jones et al. (2020) (Supplementary Information).

Microplastics analysis

Concerns about sample size for the MP extraction led to pooling two halves of one-centimeter layers, which resulted in reducing the resolution of the chronology but increasing the dry weight per sample (~20 g d.w.). The extraction and identification of MP particles for each sample was performed applying the method described in Parga Martínez et al.⁷⁴ in line with the AMAP recommendations for MP monitoring in the Arctic⁷⁵. Briefly, each sediment sample was freeze-dried overnight and treated with acetic acid 10% to dissolve foraminifera shells, KOH 5% (Potassium Hydroxide) + NaClO 7% (Sodium Hypochlorite) to digest the organic matter, and ZnCl₂ (Zinc Chloride $\delta = 1.5 \text{ g cm}^{-3}$) to separate MP particles from the inorganic fraction. The density separation step was repeated twice to extract the greatest number of small particles (extraction efficiency higher than 90% for particles >100 μm and 83% for 63–75 μm particles). Each supernatant was filtered sequentially through a customized, metal filtration unit equipped with steel filters that allows fractionation into two size classes: small (20–100 μm) and large (> 100 μm). All particles collected in both filters were chemically characterized by μFTIR imaging, that is, all particles extracted from the samples were analyzed and no sub-sampling was performed in this work. Three procedural blanks using only the solutions for MP extraction were processed simultaneously to the MP extraction of all the samples following the same protocol. The particles detected were used to calculate the limit of detection (Refer to LoD section for more details).

Quality Control

All lab equipment used during sample processing was made of glass or metal, which was muffled at 450 °C for 2 h to remove any residual plastic particles. All solutions were filtered through a muffled 0.2 μm GF/F filter ($\varnothing 47\text{mm}$), except for ZnCl₂ that was filtered through a Duran sintered disk filter funnel (pore size 10–16 μm) to avoid clogging the GF/F filter due to the high viscosity of the solution. Sample processing was conducted under a fume hood in a cleanroom laboratory with a HEPA air filter. The final filtration step was conducted under a laminar flow hood. Cotton lab coats, nitrile gloves and booties were worn during all analysis in the cleanroom. Seven air blanks were set up during sample processing to monitor background contamination in the cleanroom (Table 1).

Spectroscopy

Isolated particles from each size fraction (20–100 μm and > 100 μm), extracted from whole sediment samples, were transferred onto individual silicon membranes ($\varnothing 13 \text{ mm}$ 5 μm pore size). That is, two silicon membranes per sample. Complete membranes were analyzed via μFTIR hyperspectral imaging (Cary 670 FTIR spectrometer coupled to a Cary 620 FTIR microscope from Agilent Technologies). The microscope is equipped with a 128 × 128-pixel focal plane array (FPA) detector and all images were taken in transmission mode (32 scans, 5.5 μm pixel size, 8 cm^{-1} resolution, 3800–871 cm^{-1} spectral range). A clean silicon membrane was used as background (128 scans) prior sample measurements. The images were a square collection of 15 × 15 tiles (225 mosaic tiles), 1920 × 1920 pixels, and 761 variables (3,686,400 pixels/spectra) per sample on average. The mosaic area completely covered the filter area with all isolated particles for characterization. All data was processed using the open-source Systematic Identification of MicroPlastics in the Environment - siMPLe software (Aalborg University, Denmark and Alfred Wegener Institute, Germany), which automatically calculates a weighted average of Pearson's Correlation Coefficients between each spectrum in the mosaic area and a reference spectra library using the raw data, first and second derivatives. The algorithm applies three thresholds of probability score to reconstruct each MP particle from all the pixels of the FTIR spectra that belong to the same polymer type⁷⁶. Here, the probability thresholds were set at 0.6, 0.4 and 0.3 and a customized reference spectra library containing 94 plastic polymers and natural materials was used (Refer to Supplementary Note for details on the spectra library), which includes the top 10 specific polymers recommended for MP monitoring in the Arctic⁷⁵. Polyamide was excluded from the analysis to avoid false positive matches with protein.

The dataset provided polymer type, particle count, size as the major dimension (μm), and estimated mass (ng). Mass is calculated on siMPLe from the volume of an assumed ellipsoid morphology and the polymer density of 67% estimated thickness of the minor dimension⁷⁷. Therefore, mass is an estimated value and a general trend (mg Kg^{-1}) is presented in Supplementary Table 1. The results are shown in number of particles per kilogram of sediment (particles Kg^{-1}) normalized to dry weight. The MP burial rate (particles $\text{m}^{-2} \text{y}^{-1}$) was calculated based on particle count and sedimentation rate.

Limit of detection (LoD)

The limit of detection was calculated based on the total particles detected and characterized in the procedural blanks ($n = 3$), which were run in parallel during the MP extraction and analyzed in the same manner as described above. The LoD was determined for the mean number of MP particles and per polymer type identified in the procedural blanks: mean blank + 3 × SD blank⁷⁸. Values from procedural blanks are not subtracted from the results as it is recommended that the blank results for sediment and sand be reported uncorrected to reflect the efficiency of the precautionary methods during sample processing in the laboratory⁷⁹. Table 2 summarizes the results of the procedural blanks.

Statistical analysis

Significant differences in MP accumulation, MP Burial Rate and size fractions between Plastic Boom periods was determined using a Wilcoxon Test. Spearman's correlation was tested between Sedimentation Rate, MP Burial Rate, water content and porosity against time and the depth profile. The Shannon-Wiener Diversity Index was applied to the overall presence of polymer across time.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets generated during the current study are publically available from Pangea repository at <https://doi.org/10.1594/PANGAEA.972635>.

Particle size dataset is available at <https://doi.org/10.1594/PANGAEA.972712>. Particle morphology dataset is available at <https://doi.org/10.1594/PANGAEA.972713>. The limit of detection data is available at <https://doi.org/10.1594/PANGAEA.972709>

Received: 31 March 2024; Accepted: 4 October 2024;
Published online: 11 October 2024

References

- Steffen, W., Crutzen, P. & McNeill, J. R. The Anthropocene: are humans now overwhelming the great forces of nature. *AMBIO: A Journal of the Human Environment* **36**, 614–621 (2007).
- Waters, C. N. et al. Global boundary stratotype section and point (GSSP) for the Anthropocene series: where and how to look for potential candidates. *Earth Sci Rev* **178**, 379–429 (2018).
- Waters, C. N. et al. The Anthropocene is functionally and stratigraphically distinct from the Holocene. *Science* **351**, 1979 (2016).
- Martin, J., Lusher, A. L. & Nixon, F. C. A review of the use of microplastics in reconstructing dated sedimentary archives. *Science of The Total Environment* **806**, 150818 (2022).
- Martin, J. et al. The power of multi-matrix monitoring in the Pan-Arctic region: plastics in water and sediment. *Arct Sci* **9**, 146–164 (2023).
- Andersen, T. J. Some Practical Considerations Regarding the Application of ^{210}Pb and ^{137}Cs Dating to Estuarine Sediments. in *Applications of Paleoenvironmental Techniques in Estuarine Studies* (eds. Weckstrom, K., Saunders, K., Gell, P. & Skilbeck, C.) 121–140 (Springer, 2017). https://doi.org/10.1007/978-94-024-0990-1_6.
- Dahl, M. et al. A temporal record of microplastic pollution in Mediterranean seagrass soils. *Environmental Pollution* **273**, 116451 (2021).
- Dong, M. et al. The rapid increases in microplastics in urban lake sediments. *Sci Rep* **10**, 848 (2020).
- Fan, Y., Zheng, K., Zhu, Z., Chen, G. & Peng, X. Distribution, sedimentary record, and persistence of microplastics in the Pearl River catchment, China. *Environmental Pollution* **251**, 862–870 (2019).
- Martin, C. et al. Exponential increase of plastic burial in mangrove sediments as a major plastic sink. *Sci Adv* **6**, 1–8 (2020).
- Turner, S., Horton, A. A., Rose, N. L. & Hall, C. A temporal sediment record of microplastics in an urban lake, London, UK. *J Paleolimnol* <https://doi.org/10.1007/s10933-019-00071-7> (2019).
- Chen, M. et al. Forty-year pollution history of microplastics in the largest marginal sea of the western Pacific. *Geochem Perspect Lett* 42–47 <https://doi.org/10.7185/geochemlet.2012> (2020).
- Courtene-Jones, W., Quinn, B., Ewins, C., Gary, S. F. & Narayanaswamy, B. E. Microplastic accumulation in deep-sea sediments from the Rockall Trough. *Mar Pollut Bull* **154**, 111092 (2020).
- Tsuchiya, M. et al. Distribution of microplastics in bathyal- to hadal-depth sediments and transport process along the deep-sea canyon and the Kuroshio Extension in the Northwest Pacific. *Mar Pollut Bull* 115466 (2023) <https://doi.org/10.1016/j.marpolbul.2023.115466>.
- Uddin, S. et al. A review of microplastic distribution in sediment profiles. *Mar Pollut Bull* **163**, 111973 (2021).
- Xue, B. et al. Underestimated microplastic pollution derived from fishery activities and “hidden” in deep sediment. *Environ Sci Technol* **54**, 2210–2217 (2020).
- Adams, J. K. et al. Anthropogenic particles (including microfibrers and microplastics) in marine sediments of the Canadian Arctic. *Science of The Total Environment* **784**, 147155 (2021).
- Bergmann, M. et al. White and wonderful? Microplastics prevail in snow from the Alps to the Arctic. *Sci Adv* **5**, eaax1157 (2019).
- Kanhai, L. D. K. et al. Microplastics in sub-surface waters of the Arctic Central Basin. *Mar Pollut Bull* **130**, 8–18 (2018).
- Peeken, I. et al. Arctic sea ice is an important temporal sink and means of transport for microplastic. *Nat Commun* **9**, 1505 (2018).
- Eisted, R. & Christensen, T. H. Waste management in Greenland: current situation and challenges. *Waste Management & Research: The Journal for a Sustainable Circular Economy* **29**, 1064–1070 (2011).
- Gunnarsdóttir, R., Jenssen, P. D., Erland Jensen, P., Villumsen, A. & Kallenborn, R. A review of wastewater handling in the Arctic with special reference to pharmaceuticals and personal care products (PPCPs) and microbial pollution. *Ecol Eng* **50**, 76–85 (2013).
- Departementet for Forskning og Miljø. *Affaldshandlingsplan 2020–2031*. (2020). <https://faolex.fao.org/docs/pdf/grl201496.pdf>.
- Haarr, M. L. et al. Beach litter sources around Nuuk, Greenland: An analysis by UArctic summer school graduate course students. *Mar Pollut Bull* **191**, 114914 (2023).
- Mallory, M. L. et al. Anthropogenic litter in marine waters and coastlines of Arctic Canada and West Greenland. *Science of The Total Environment* **783**, 146971 (2021).
- Syberg, K. et al. A nationwide assessment of plastic pollution in the Danish realm using citizen science. *Sci Rep* **10**, 17773 (2020).
- Statistics Greenland. Greenland in figures 2023. *Statistics Greenland* www.stat.gl (2023).
- Mikkelsen, N., Ingerslev, T. & (eds). *Nomination of the Ilulissat Icefjord for Inclusion in the World Heritage List*. Geological Survey of Denmark and Greenland (GEUS) (2002).
- Amélineau, F. et al. Microplastic pollution in the Greenland Sea: Background levels and selective contamination of planktivorous diving seabirds. *Environmental Pollution* **219**, 1131–1139 (2016).
- Liboiron, M. et al. Abundance and types of plastic pollution in surface waters in the Eastern Arctic (Inuit Nunangat) and the case for reconciliation science. *Science of The Total Environment* **782**, 146809 (2021).
- Morgana, S. et al. Microplastics in the Arctic: a case study with sub-surface water and fish samples off Northeast Greenland. *Environmental Pollution* **242**, 1078–1086 (2018).
- Jiang, Y., Yang, F., Zhao, Y. & Wang, J. Greenland Sea Gyre increases microplastic pollution in the surface waters of the Nordic Seas. *Science of The Total Environment* **712**, 136484 (2020).
- Rist, S. et al. Quantification of plankton-sized microplastics in a productive coastal Arctic marine ecosystem. *Environmental Pollution* **266**, 115248 (2020).
- Cózar, A. et al. The Arctic Ocean as a dead end for floating plastics in the North Atlantic branch of the Thermohaline Circulation. *Sci Adv* **3**, e1600582 (2017).
- Tekman, M. B. et al. Tying up Loose Ends of Microplastic Pollution in the Arctic: Distribution from the Sea Surface through the Water Column to Deep-Sea Sediments at the HAUSGARTEN Observatory. *Environ Sci Technol* **54**, 4079–4090 (2020).
- Appleby, P. G. Chronostratigraphic Techniques in Recent Sediments. in *Tracking Environmental Change Using Lake Sediments* (ed. Last, W. M., Smol, J. P.) 171–203 (Kluwer Academic Publishers, Dordrecht, 2002) https://doi.org/10.1007/0-306-47669-X_9
- Streuff, K. et al. Seafloor geomorphology and glaciomarine sedimentation associated with fast-flowing ice sheet outlet glaciers in Disko Bay, West Greenland. *Quat Sci Rev* **169**, 206–230 (2017).
- Clough, L. M. et al. Infaunal density, biomass and bioturbation in the sediments of the Arctic Ocean. *Deep Sea Research Part II: Topical Studies in Oceanography* **44**, 1683–1704 (1997).
- Soltwedel, T. et al. Bioturbation rates in the deep Fram Strait: Results from in situ experiments at the arctic LTER observatory HAUSGARTEN. *J Exp Mar Biol Ecol* **511**, 1–9 (2019).
- Buch, E. *Present Oceanographic Conditions in Greenland Waters*. <https://www.dmi.dk/fileadmin/Rapporter/SR/sr02-02.pdf> (2000).
- Kim, S.-K. et al. Importance of seasonal sea ice in the western Arctic ocean to the Arctic and global microplastic budgets. *J Hazard Mater* **418**, 125971 (2021).
- Obbard, R. W. et al. Global warming releases microplastic legacy frozen in Arctic Sea ice. *Earths Future* **2**, 315–320 (2014).

43. Lusher, A. L., Tirelli, V., O'Connor, I. & Officer, R. Microplastics in Arctic polar waters: the first reported values of particles in surface and sub-surface samples. *Sci Rep* **5**, 14947 (2015).
44. Zhang, Y. et al. Cryosphere as a temporal sink and source of microplastics in the Arctic region. *Geoscience Frontiers* **14**, 101566 (2023).
45. Isfjordscentret. *Kangiata Illorsua - Ilulissat Icefjord Centre*. (2021).
46. Zalasiewicz, J. et al. The geological cycle of plastics and their use as a stratigraphic indicator of the Anthropocene. *Anthropocene* **13**, 4–17 (2016).
47. Wang, F. et al. Multiple dating approaches applied to the recent sediments in the Yangtze River (Changjiang) subaqueous delta. *Holocene* **28**, 858–866 (2018).
48. Simon-Sánchez, L. et al. Can a Sediment Core Reveal the Plastic Age? Microplastic Preservation in a Coastal Sedimentary Record. *Environ Sci Technol* **56**, 16780–16788 (2022).
49. Ramasamy, E. V., Sruthy, S., Harit, A. K., Mohan, M. & Binish, M. B. Microplastic pollution in the surface sediment of Kongsfjorden, Svalbard, Arctic. *Mar Pollut Bull* **173**, 112986 (2021).
50. Fang, C. et al. Spatio-temporal variation of microplastic pollution in the sediment from the Chukchi Sea over five years. *Science of The Total Environment* **806**, 150530 (2022).
51. Kim, S.-K. et al. Arctic Ocean sediments as important current and future sinks for marine microplastics missing in the global microplastic budget. *Sci Adv* **9**, 1–15 (2023).
52. Zhang, T. et al. Accumulation of microplastics in the marine sediments of the Chukchi Sea, Arctic Ocean. *Reg Stud Mar Sci* **70**, 103363 (2024).
53. Dris, R., Gasperi, J., Saad, M., Mirande, C. & Tassin, B. Synthetic fibers in atmospheric fallout: A source of microplastics in the environment? *Mar Pollut Bull* **104**, 290–293 (2016).
54. Strand, J., Bach, L. & Juul-Pedersen, T. Shoreline litter monitoring surveys in Greenland 2016 – 2022. *Second International Symposium on Plastics in the Arctic and sub-Arctic Region* (2023).
55. Hendriksen, K. & Hoffmann, B. Greenlandic water and sanitation—a context oriented analysis of system challenges towards local sustainable development. *Environmental Science and Pollution Research* **25**, 33014–33024 (2018).
56. Bond, T., Ferrandiz-Mas, V., Felipe-Sotelo, M. & van Sebille, E. The occurrence and degradation of aquatic plastic litter based on polymer physicochemical properties: A review. *Crit Rev Environ Sci Technol* **48**, 685–722 (2018).
57. OSPAR. Scoping Study to identify key waste items from the fishing industry and aquaculture. *Marine Litter Regional Action Plan Action 35* <https://www.ospar.org/documents?v=41242> (2019).
58. Papineschi, J., Hogg, D., Chowdhury, T., Durrant, C. & Thomson, A. *Analysis of Nordic Regulatory Framework and Its Effect on Waste Prevention and Recycling in the Region*. (Nordic Council of Ministers, Copenhagen, 2019). <https://doi.org/10.6027/TN2019-522>.
59. Andrady, A. L. Microplastics in the marine environment. *Mar Pollut Bull* **62**, 1596–1605 (2011).
60. Law, K. L. et al. Plastic accumulation in the North Atlantic subtropical gyre. *Science* (1979) **329**, 1185–1188 (2010).
61. van Sebille, E. et al. A global inventory of small floating plastic debris. *Environmental Research Letters* **10**, 124006 (2015).
62. Porter, A., Lyons, B. P., Galloway, T. S. & Lewis, C. Role of marine snows in microplastic fate and bioavailability. *Environ Sci Technol* **52**, 7111–7119 (2018).
63. Booth, A. M. & Sørensen, L. Microplastic Fate and Impacts in the Environment. in *Handbook of Microplastics in the Environment* 1–24 (Springer International Publishing, Cham, 2020). https://doi.org/10.1007/978-3-030-10618-8_29-1.
64. Laursen, S. N. et al. Settling of buoyant microplastic in estuaries: The importance of flocculation. *Science of The Total Environment* **886**, 163976 (2023).
65. Bråte, I. L. N. et al. Mytilus spp. as sentinels for monitoring microplastic pollution in Norwegian coastal waters: A qualitative and quantitative study. *Environmental Pollution* **243**, 383–393 (2018).
66. Loughlin, C., Marques Mendes, A. R., Morrison, L. & Morley, A. The role of oceanographic processes and sedimentological settings on the deposition of microplastics in marine sediment: Icelandic waters. *Mar Pollut Bull* **164**, 111976 (2021).
67. Lorenz, C. et al. Spatial distribution of microplastics in sediments and surface waters of the southern North Sea. *Environmental Pollution* **252**, 1719–1729 (2019).
68. Bergmann, M. et al. High quantities of microplastic in Arctic deep-sea sediments from the HAUSGARTEN observatory. *Environ Sci Technol* **51**, 11000–11010 (2017).
69. Rogers, K. L., Carreres-Calabuig, J. A., Gorokhova, E. & Posth, N. R. Micro-by-micro interactions: How microorganisms influence the fate of marine microplastics. *Limnol Oceanogr Lett* **5**, 18–36 (2020).
70. Worm, B., Lotze, H. K., Jubinville, I., Wilcox, C. & Jambeck, J. Plastic as a persistent marine pollutant. *Annu Rev Environ Resour* **42**, 1–26 (2017).
71. ECHA. *Opinion on an Annex XV Dossier Proposing Restrictions on Intentionally-Added Microplastics*. Committee of Risk Assessment (RAC) and Committee for Socio-economic Analysis (SEAC) (2020).
72. Gilbert, R., Nielsen, N., Desloges, J. R. & Rasch, M. Contrasting glaciomarine sedimentary environments of two arctic fiords on Disko, West Greenland. *Mar Geol* **147**, 63–83 (1998).
73. Boertmann, D., Mosbech, A., Schiedek, D. & Dünweber, M. *Disko West. A Strategic Environmental Impact Assessment of Hydrocarbon Activities*. <https://dce2.au.dk/pub/SR71.pdf> (2013).
74. Parga Martinez, K. B., da Silva, V. H., Andersen, T. J., Posth, N. R. & Strand, J. Improved separation and quantification method for microplastic analysis in sediment: a fine-grained matrix from Arctic Greenland. *Mar Pollut Bull* **196**, 115574 (2023).
75. Primpke, S. et al. Monitoring of microplastic pollution in the Arctic: recent developments in polymer identification, quality assurance and control, and data reporting. *Arct Sci* **9**, 176–197 (2023).
76. Primpke, S. et al. EXPRESS: Toward the Systematic Identification of Microplastics in the Environment: Evaluation of a New Independent Software Tool (siMPle) for Spectroscopic Analysis. *Appl Spectrosc* **000370282091776** (2020) <https://doi.org/10.1177/0003702820917760>.
77. Simon, M., van Alst, N. & Vollertsen, J. Quantification of microplastic mass and removal rates at wastewater treatment plants applying Focal Plane Array (FPA)-based Fourier Transform Infrared (FT-IR) imaging. *Water Res* **142**, 1–9 (2018).
78. MacDougall, D. et al. Guidelines for data acquisition and data quality evaluation in environmental chemistry. *Anal Chem* **52**, 2242–2249 (1980).
79. Galgani, F. et al. *Guidance on the Monitoring of Marine Litter in European Seas*. <https://publications.jrc.ec.europa.eu/repository/handle/JRC83985> (2023) <https://doi.org/10.2788/99475>.

Acknowledgements

This work has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie (grant agreement No 801199), the VILLUM Young Investigator Programme (grant 15397), the Danmarks Frie Forskningsfond FNU (grant 11-105806), and the Danish Center for Research on Plastic Pollution (MarinePlastic) via the Velux Foundations. The authors thank Jens Lykke Borch from Greenland's Fisheries License Control Authority (GFLK) for providing data on fishing effort in Disko Bay, and Andreas Xion Magnussen Griis from the Avannaata Municipality for providing data on waste collection at the Ilulissat Incineration Plant. No permission were required to access the sampling site.

Author contributions

KPM conceptualization, data curation, funding acquisition, formal analysis, methodology, visualization, writing – original draft, review & editing; TJA

funding acquisition, resources, supervision, writing – review & editing; VHS investigation, methodology, supervision, writing – review & editing; JS funding acquisition, methodology, supervision, writing – review & editing; NRP funding acquisition, supervision, writing – review & editing.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s43247-024-01768-y>.

Correspondence and requests for materials should be addressed to Karla Parga Martínez.

Peer review information *Communications Earth & Environment* thanks Kathryn Willis, Laura Simon-Sánchez and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Clare Davis and Alice Drinkwater. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024